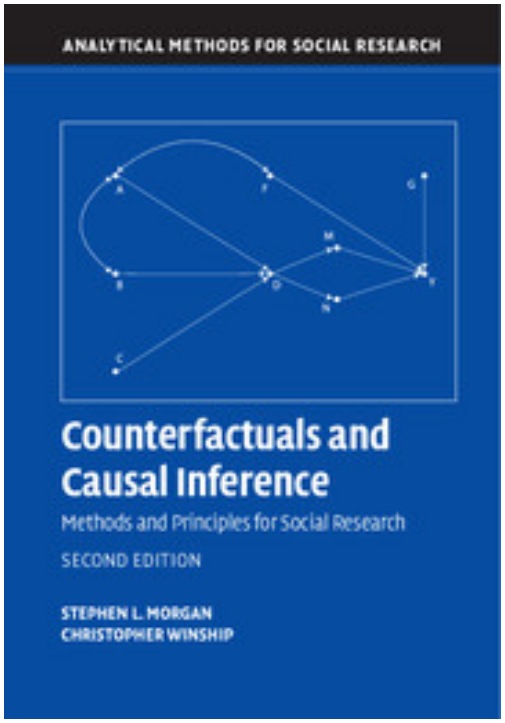
# Counterfactuals and Causal Inference

## Methods and Principles for Social Research




Chapter 7

# Outline


1. Weighted Regression Estimators of the ATE
2. Weighted Regression Estimators of the ATT and the ATC
3. Doubly Robust Weighted Regression Estimators
4. Remaining Practical Issues in Weighted Regression Analysis
5. An Extended Example of Weighted Regression

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1. Weighted Regression Estimators of the ATE
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# Inverse probability weighting

- To estimate the ATE with weighted regression, one first must estimate the predicted probability of treatment, $\hat{p}_i$
- Once the values for $\hat{p}_i$ are obtained, weights for the ATE are constructed as

$$\text{For } d_i = 1: \quad w_{i,\,\text{ATE}} = \frac{1}{\hat{p}_i}, \tag{7.1}$$

$$\text{For } d_i = 0: \quad w_{i,\,\text{ATE}} = \frac{1}{1-\hat{p}_i}.$$

# Weighted regression estimation

- To then estimate the ATE, a weighted bivariate regression model is estimated, where $y_i$ are the values for the outcome, $d_i$ are the values for the sole predictor variable, and $w_{i,\,\mathrm{ATE}}$ are the weights.
- No specialized software is required, and the weights are treated exactly as if they are survey weights (even though they are not survey weights, but instead weights based on estimated propensity scores that are relevant only for the estimation of the ATE).

- For those accustomed to matrix representations of regression estimates, note first that the naive estimator can be written as an OLS estimator, $(\mathbf{Q}'\mathbf{Q})^{-1}\mathbf{Q}'\mathbf{y}$
- To estimate the ATE, a weighted regression estimator is utilized,

$$\hat{\delta}_{\text{OLS, weighted}} \equiv (\mathbf{Q}'\mathbf{P}\mathbf{Q})^{-1}\mathbf{Q}'\mathbf{P}\mathbf{y}, \tag{7.2}$$

  where $\mathbf{P}$ is an $n \times n$ diagonal matrix with the corresponding value of $w_{i,\ \text{ATE}}$ on the diagonal for each individual and zero elsewhere.

# Weighted Regression Demonstration 1

Setup for Variant 1: As for Matching Demonstration 3, the potential outcomes were specified as functions of individual values for $A$ and $B$:

$$y_i^1 = 102 + 6a_i + 4b_i + \upsilon_i^1, \tag{7.3}$$
$$y_i^0 = 100 + 3a_i + 2b_i + \upsilon_i^0, \tag{7.4}$$

where $A$ and $B$ are distributed as independent uniform random variables with a minimum of .1 and a maximum of 1, and where $\upsilon_i^1$ and $\upsilon_i^0$ are independent random draws from a normal distribution with expectation 0 and a standard deviation of 5.

Setup for Variant 2: Equations (7.3) and (7.4) are replaced with

$$y_i^1 = 102 + 3a_i + 2b_i + 6(a_i \times b_i) + \upsilon_i^1, \tag{7.5}$$

$$y_i^0 = 100 + 2a_i + 1b_i - 2(a_i \times b_i) + \upsilon_i^0, \tag{7.6}$$

**Table 7.1** Weighted Regression Estimates of the ATE, Using and Extending the Data Setup for Matching Demonstration 3

| | ATE |
|---|---|
| Variant I: $Y^1$ and $Y^0$ linear in $A$ and $B$ | |
| True treatment effect | 4.53 |
| OLS regression estimates: | |
| $Y$ regressed on $D$ | 5.39 |
| $Y$ regressed on $D$ and linear $A$ and $B$ | 4.75 |
| $Y$ regressed on $D$ and quadratic $A$ and $B$ | 4.74 |
| ATE-weighted regression of $Y$ on $D$ | 4.53 |
| Variant II: $Y^1$ and $Y^0$ nonlinear in $A$ and $B$ | |
| True treatment effect | 5.05 |
| OLS regression estimates: | |
| $Y$ regressed on $D$ | 5.88 |
| $Y$ regressed on $D$ and linear $A$ and $B$ | 5.47 |
| $Y$ regressed on $D$ and quadratic $A$ and $B$ | 5.44 |
| ATE-weighted regression of $Y$ on $D$ | 5.05 |

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# Inverse probability weighting again

To estimate the ATT and ATC, two sets of weights $w_{i,\,\text{ATT}}$ and $w_{i,\,\text{ATC}}$ are formed as

$$\begin{aligned} &\text{For } d_i = 1: \quad w_{i,\,\text{ATT}} = 1, \\ &\text{For } d_i = 0: \quad w_{i,\,\text{ATT}} = \frac{\hat{p}_i}{1-\hat{p}_i}, \end{aligned} \tag{7.7}$$

and

$$\begin{aligned} &\text{For } d_i = 1: \quad w_{i,\,\text{ATC}} = \frac{1-\hat{p}_i}{\hat{p}_i}, \\ &\text{For } d_i = 0: \quad w_{i,\,\text{ATC}} = 1. \end{aligned} \tag{7.8}$$

# Weighted Regression Demonstration 1 (cont'd)

**Table 7.2** Weighted Regression Estimates of the ATT and ATC, Using and Extending the Data Setup for for Matching Demonstration 3

| | ATT | ATC |
|---|---|---|
| Variant I: $Y^1$ and $Y^0$ linear in $A$ and $B$ | | |
| True treatment effects | 4.89 | 4.40 |
| OLS regression estimates: | | |
| $Y$ regressed on $D$ | 5.39 | 5.39 |
| $Y$ regressed on $D$ and linear $A$ and $B$ | 4.75 | 4.75 |
| $Y$ regressed on $D$ and quadratic $A$ and $B$ | 4.74 | 4.74 |
| ATT or ATC weighted regression of $Y$ on $D$ | 4.89 | 4.40 |
| Variant II: $Y^1$ and $Y^0$ nonlinear in $A$ and $B$ | | |
| True treatment effects | 5.77 | 4.79 |
| OLS regression estimates: | | |
| $Y$ regressed on $D$ | 5.88 | 5.88 |
| $Y$ regressed on $D$ and linear $A$ and $B$ | 5.47 | 5.47 |
| $Y$ regressed on $D$ and quadratic $A$ and $B$ | 5.44 | 5.44 |
| ATT or ATC weighted regression of $Y$ on $D$ | 5.77 | 4.79 |

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# Doubly robust and doubly protected

- Recall Equation (7.2),

$$\hat{\delta}_{\text{OLS, weighted}} \equiv (\mathbf{Q}'\mathbf{P}\mathbf{Q})^{-1}\mathbf{Q}'\mathbf{P}\mathbf{y},$$

  where we defined the $\mathbf{Q}$ matrix as containing a column of $1s$ and a column of individual-level values $d_i$ for the treatment variable, along with a diagonal matrix $\mathbf{P}$ that represents a set of weights specific to the target parameter.
- For this estimator, as used up until this section, treatment assignment variables are presumed to have been used only in a first stage to estimate the values of $\hat{p}_i$, which then structure the appropriate weights in $\mathbf{P}$.
- Doubly robust estimators use the same variables a second time as supplementary adjustment variables. The estimator in Equation (7.2) is still used, but these variables are included in additional columns in $\mathbf{Q}$.

# Weighted Regression Demonstration 2

**Table 7.3** Bias for Weighted Regression Estimates of the ATT, Catholic Schooling on Achievement Across 10 Simulated Datasets Utilized for Matching Demonstration 4 and Regression Demonstration 4

| | Specification of treatment assignment variables: | | | |
|---|---|---|---|---|
| | Incomplete specification | | Complete specification | |
| Method | Min, Max | Average | Min, Max | Average |
| Weighted regression estimate of the ATT | -0.08, 2.17 | 0.82 | -0.87, 1.11 | -0.09 |
| Doubly robust weighted regression estimate of the ATT | -0.05, 2.19 | 0.83 | -0.87, 1.08 | -0.10 |

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# Extreme weights

- Extreme weights may be correct but still problematic. How to deal with this situation is a classic tradeoff between bias and variance.
- In other cases, the weights may not be correct. In this case, they are more likely to have too little variance.
- Various types of "trimming" should be considered if some weights appear unusually large.

# Survey data

- First, weight the propensity-score-estimating equation by the appropriate survey weight suggested by the data distributor.
- Multiply the constructed weights for the ATE, ATT, and/or ATC by the same survey weight. In so doing, the analyst then passes to the regression routine a model-based weight for the relevant parameter that is modified by the probability of inclusion in the analysis sample that is being utilized.
- For example, the ATE weights can be formed as

$$\text{For } d_i = 1: \quad w_{i,\ \text{ATE, complex}} = \text{survey weight} \ \times \ \frac{1}{\hat{p}_i}, \qquad (7.9)$$

$$\text{For } d_i = 0: \quad w_{i,\ \text{ATE, complex}} = \text{survey weight} \ \times \ \frac{1}{1-\hat{p}_i}.$$

# Expected sampling variance and estimated standard errors

- No convincing rationale exists for assuming homoscedasticity when calculating standard errors. We recommend that researchers routinely report heteroscedasticity-consistent standard errors (e.g., the *robust* option in Stata, which will be called automatically if the weights are specified as *pweights*).
- These estimated standard errors are more likely to reflect the true expected sampling variability of weighted regression estimates, although the degree to which robust variance estimation is fully effective in finite samples (especially small ones) has not been clearly established in the literature.
- In our own simulation work, we have found that heteroscedasticity-consistent standard errors have performed quite well for larger samples and the estimation of parameters that are identified.

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# Weighted Regression Demonstration 3

The data: the 2002 base-year and 2004 follow-up waves of the Education Longitudinal Study (ELS)

- A nationally representative sample of students in public and private high schools, based on a two-stage sampling design that first draws a random sample of public and private high schools and then draws random within-school samples of tenth graders (typically aged 16).
- For the first follow-up in 2004, respondents were tracked to alternative destinations, and most respondents were twelfth graders (typically aged 18).
- We restricted the analysis to respondents who were enrolled in either a Catholic school or a public school during the 2001–02 academic school year.

**Table 7.4** Means and Standard Deviations of the Primary Variables Used in the Demonstration

| Variable | Public Mean | Public SD | Catholic Mean | Catholic SD |
|---|---|---|---|---|
| Math Test Scores | | | | |
| IRT estimated number right (10th grade) | 41.68 | 13.97 | 48.99 | 12.02 |
| IRT estimated number right (12th grade) | 47.64 | 15.05 | 56.08 | 12.80 |
| Math gain (12th grade – 10th grade) | 4.66 | 6.49 | 6.66 | 6.06 |
| Female | .50 | | .48 | |
| Race (White is the reference category) | | | | |
| Black | .15 | | .06 | |
| Hispanic | .17 | | .11 | |
| Asian | .04 | | .04 | |
| Native American | .01 | | <.01 | |
| Multiracial | .04 | | .04 | |
| Urbanicity (Suburban is the reference category) | | | | |
| Urban | .28 | | .58 | |
| Rural | .21 | | .01 | |
| Region (Midwest is the reference category) | | | | |
| Northeast | .18 | | .31 | |
| South | .34 | | .23 | |
| West | .23 | | .17 | |
| Family Background | | | | |
| Mother's education (in years) | 13.46 | 2.32 | 14.77 | 2.22 |
| Father's education (in years) | 13.59 | 2.59 | 15.25 | 2.57 |
| SEI score of mother's occupation | 44.98 | 12.87 | 50.55 | 12.85 |
| SEI score of father's occupation | 44.15 | 11.70 | 49.81 | 11.71 |
| Family income (natural log) | 10.60 | 1.09 | 11.23 | .90 |
| Family income (natural log) squared | 113.61 | 19.83 | 126.99 | 17.04 |
| Family income (natural log) cubed | 1225.64 | 295.46 | 1441.33 | 267.98 |
| Two-parent family | .75 | | .84 | |
| Past History (as reported by parent) | | | | |
| Learning disability | .13 | | .07 | |
| Ever held back | .13 | | .05 | |
| Repeated 4th grade | .01 | | <.01 | |
| Years parents lived in current neighborhood | 10.56 | 8.00 | 12.90 | 8.21 |

*Source*: Education Longitudinal Study, 2002 and 2004

# Steps 1 and 2: Estimate regression models

- We first estimate three bivariate regression equations with ordinary least squares,

$$Y = \hat{\alpha} + \hat{\delta}_{\text{OLS, bivariate}} D + \varepsilon, \tag{7.10}$$

  where $Y$ is one of three interval-scaled outcome variables and $D$ is an indicator variable equal to 1 for those who attend Catholic high schools (and equal to 0 for those who attend public high schools).
- The estimated coefficient $\hat{\delta}_{\text{OLS, bivariate}}$ is the estimated effect of $D$ on $Y$.

- We next estimate three multiple regression equations with ordinary least squares,

$$Y = \hat{\alpha} + \hat{\delta}_{\text{OLS, multiple}} D + X\hat{\beta} + \varepsilon, \tag{7.11}$$

where $X$ represents observed variables thought to determine $D$ and $Y$ (because they lie on back-door paths from $D$ to $Y$), $\hat{\delta}_{\text{OLS, multiple}}$ is the estimated causal effect of $D$ on $Y$ adjusted for $X$, and $\hat{\beta}$ is a conformable vector of estimated coefficients that correspond to the variables in $X$.

**Table 7.5** Catholic School Coefficients from Baseline Regression Models Predicting Tenth Grade Math Test Scores, Twelfth Grade Math Test Scores, and Math Test Gains

| Predictor Variables | Outcome Variable: 10th Grade Math Test Score |
|---|---|
| Model 1: Dummy for Catholic school | 7.31 |
| | (.66) |
| Model 2: Model 1 + family background, demographics, and past history | 1.48 |
| | (.52) |
| | **Outcome Variable: 12th Grade Math Test Score** |
| Model 1: Dummy for Catholic school | 8.45 |
| | (.74) |
| Model 2: Model 1 + family background, demographics, and past history | 2.13 |
| | (.60) |
| | **Outcome Variable: Math Gain (12th – 10th)** |
| Model 1: Dummy for Catholic school | 2.00 |
| | (.23) |
| Model 2: Model 1 + family background, demographics, and past history | 1.27 |
| | (.26) |

*Notes*: Heteroscedasticity-consistent standard errors in parentheses.

# Step 3: Estimate propensity scores and form weights to estimate the ATT and ATC

- We iteratively estimate propensity scores in pursuit of maximum achievable balance on the variables in $X$ and then form weights based on the final set of estimates.
- We first estimated a model of treatment selection/assignment using the variables in $X$ as predictors, and using the same simple linear specification as for the multiple regression models estimated for Table 7.5.

# Along the way, we assess balance

- The standardized difference of the mean for each variable in $X$ is calculated as

$$\frac{|E_N[x_i|d_i = 1] - E_N[x_i|d_i = 0]|}{\sqrt{\frac{1}{2}\mathrm{Var}_N[x_i|d_i = 1] + \frac{1}{2}\mathrm{Var}_N[x_i|d_i = 0]}} \tag{7.12}$$

- An analog for standardized differences in standard deviations (although statisticians like to use ratios of variances, etc.)

# For the simple linear specification

- The results showed that the weights succeeded in producing substantial balance, reducing the average standardized difference of means from .350 to .00634 when using $w_{i,\ \mathrm{ATT}}$ and to .111 when using $w_{i,\ \mathrm{ATC}}$.
- The average standardized difference of standard deviations also fell substantially from .0715 to .0391 when using $w_{i,\ \mathrm{ATT}}$ and to .0287 when using $w_{i,\ \mathrm{ATC}}$.
- The increase in balance that results from employing $w_{i,\ \mathrm{ATT}}$ and $w_{i,\ \mathrm{ATC}}$ is substantial, but some minor imbalance remains.

# With an enriched specification

- Traditional trial and error respecification, resulting in an additional 75 interaction terms
- The average standardized difference of means fell further from .00634 to .00437 when using $w_{i,\ \mathrm{ATT}}$ and from .111 to .0899 when using $w_{i,\ \mathrm{ATC}}$.
- Likewise, the average standardized difference of standard deviations also fell from .0391 to .0166 when using $w_{i,\ \mathrm{ATT}}$ and from .0287 to .0229 when using $w_{i,\ \mathrm{ATC}}$.
- By the standards in this literature, this very good balance.

**Table 7.6** Means and Standard Deviations of Primary Variables, Weighted by the ATT Weight from the Final Estimation of the Treatment Assignment Model

| | Public | | Catholic | |
|---|---|---|---|---|
| Variable | Mean | SD | Mean | SD |
| Female | .48 | | .48 | |
| Race (White is the reference category) | | | | |
| Black | .06 | | .06 | |
| Hispanic | .11 | | .11 | |
| Asian | .04 | | .04 | |
| Native American | <.01 | | <.01 | |
| Multiracial | .04 | | .04 | |
| Urbanicity (Suburban is the reference category) | | | | |
| Urban | .58 | | .58 | |
| Rural | .01 | | .01 | |
| Region (Midwest is the reference category) | | | | |
| Northeast | .31 | | .31 | |
| South | .23 | | .23 | |
| West | .16 | | .17 | |
| Family Background | | | | |
| Mother's education (in years) | 14.77 | 2.22 | 14.77 | 2.22 |
| Father's education (in years) | 15.25 | 2.57 | 15.25 | 2.57 |
| SEI score of mother's occupation | 50.56 | 12.77 | 50.55 | 12.85 |
| SEI score of father's occupation | 49.67 | 11.68 | 49.81 | 11.71 |
| Family income (natural log) | 11.24 | .84 | 11.23 | .90 |
| Family income (natural log) squared | 127.07 | 16.50 | 126.99 | 17.04 |
| Family income (natural log) cubed | 1441.92 | 261.94 | 1441.33 | 267.98 |
| Two-parent family | .83 | | .84 | |
| Past History (as reported by parent) | | | | |
| Learning disability | .07 | | .07 | |
| Ever held back | .05 | | .05 | |
| Repeated 4th grade | <.01 | | <.01 | |
| Years parents lived in current neighborhood | 12.93 | 8.96 | 12.90 | 8.21 |

**Table 7.7** Means and Standard Deviations of Primary Variables, Weighted by the ATC Weight from the Final Estimation of the Treatment Assignment Model

| | Public | | Catholic | |
|---|---|---|---|---|
| Variable | Mean | SD | Mean | SD |
| Female | .50 | | .53 | |
| Race (White is the reference category) | | | | |
| Black | .15 | | .20 | |
| Hispanic | .17 | | .13 | |
| Asian | .04 | | .06 | |
| Native American | .01 | | .01 | |
| Multiracial | .04 | | .05 | |
| Urbanicity (Suburban is the reference category) | | | | |
| Urban | .28 | | .33 | |
| Rural | .21 | | .05 | |
| Region (Midwest is the reference category) | | | | |
| Northeast | .18 | | .28 | |
| South | .34 | | .29 | |
| West | .23 | | .16 | |
| Family Background | | | | |
| Mother's education (in years) | 13.46 | 2.32 | 13.58 | 2.35 |
| Father's education (in years) | 13.59 | 2.59 | 13.80 | 2.68 |
| SEI score of mother's occupation | 44.98 | 12.87 | 45.88 | 13.10 |
| SEI score of father's occupation | 44.15 | 11.70 | 44.05 | 11.71 |
| Family income (natural log) | 10.60 | 1.09 | 10.62 | 1.03 |
| Family income (natural log) squared | 113.61 | 19.83 | 113.87 | 19.36 |
| Family income (natural log) cubed | 1225.64 | 295.46 | 1228.94 | 292.93 |
| Two-parent family | .75 | | .71 | |
| Past History (as reported by parent) | | | | |
| Learning disability | .13 | | .14 | |
| Ever held back | .13 | | .12 | |
| Repeated 4th grade | .01 | | <.01 | |
| Years parents lived in current neighborhood | 10.56 | 8.00 | 10.72 | 6.89 |

# Lots of overlap

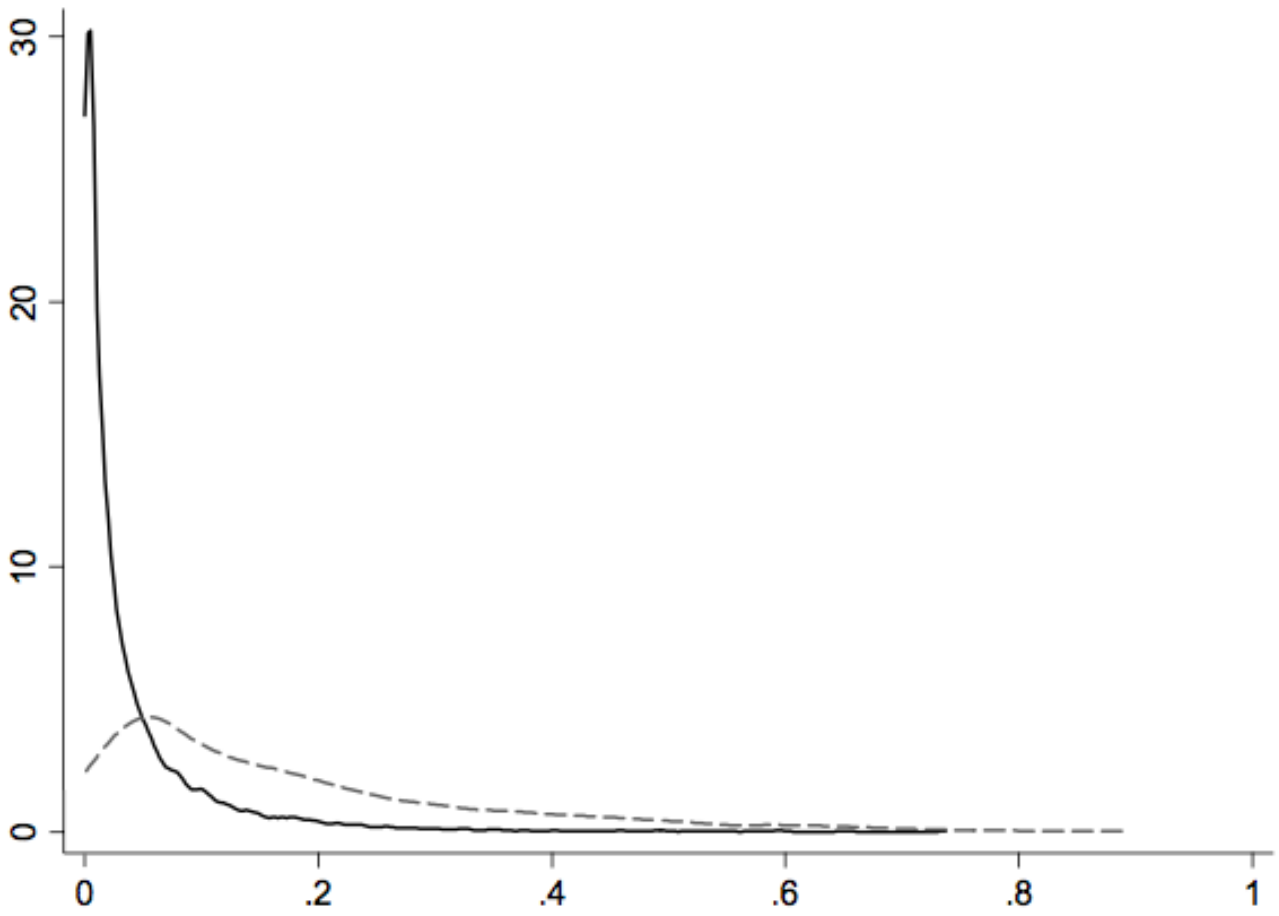


**Figure 7.1** Kernel density estimates of the estimated propensity score, calculated separately for public school students (black solid line) and Catholic school students (gray dashed line).

# Step 4: Estimate weighted regression

**Table 7.8** Catholic School Coefficients from ATT-Weighted and ATC-Weighted Regression Models Predicting Tenth Grade Math Test Scores, Twelfth Grade Math Test Scores, and Math Test Gains

| | Outcome Variable: 10th Grade Math Test Score | |
|---|---|---|
| Predictor Variables | ATT Weight | ATC Weight |
| Model 1: Dummy for Catholic school | 1.08 | 2.44 |
| | (.77) | (1.17) |
| | Outcome Variable: 12th Grade Math Test Score | |
| | ATT Weight | ATC Weight |
| Model 1: Dummy for Catholic school | 1.42 | 3.29 |
| | (.89) | (1.36) |
| | Outcome Variable: Math Gain (12th – 10th) | |
| | ATT Weight | ATC Weight |
| Model 1: Dummy for Catholic school | 1.02 | 1.98 |
| | (.31) | (.37) |

**Table 7.9** Catholic School Coefficients from Weighted Regression Models Restricted to the Region of Overlap in the Estimated Propensity Scores

| | Outcome Variable: 10th Grade Math Test Score | |
|---|---|---|
| Predictor Variables | ATT Weight | ATC Weight |
| Model 1: Dummy for Catholic school | 1.02 | 2.18 |
| | (.77) | (1.17) |
| | Outcome Variable: 12th Grade Math Test Score | |
| | ATT Weight | ATC Weight |
| Model 1: Dummy for Catholic school | 1.38 | 2.85 |
| | (.89) | (1.36) |
| | Outcome Variable: Math Gain (12th – 10th) | |
| | ATT Weight | ATC Weight |
| Model 1: Dummy for Catholic school | 1.03 | 1.82 |
| | (.31) | (.37) |

# Step 5: Estimate doubly robust weighted regression

**Table 7.10** Catholic School Coefficients from Doubly Robust Weighted Regression Models

| | Outcome Variable: 10th Grade Math Test Score | |
|---|---|---|
| Predictor Variables | ATT Weight | ATC Weight |
| Model 2: Model 1 + family background, demographics, and past history | 1.08<br>(.56) | 2.42<br>(.59) |
| | Outcome Variable: 12th Grade Math Test Score | |
| | ATT Weight | ATC Weight |
| Model 2: Model 1 + family background, demographics, and past history | 1.60<br>(.66) | 4.01<br>(.71) |
| | Outcome Variable: Math Gain (12th – 10th) | |
| | ATT Weight | ATC Weight |
| Model 2: Model 1 + family background, demographics, and past history | 1.05<br>(.29) | 2.06<br>(.45) |

# Step 6: Assess robustness for additional insight

**Table 7.11** Catholic School Coefficients from Weighted Regression Models, Including Additional Covariates

| Predictor Variables | Outcome Variable: 10th Grade Math Test Score | |
|---|---|---|
| | ATT Weight | ATC Weight |
| Model 3: Model 2 + expectations and parental involvement | .22 | 1.26 |
| | (.54) | (.62) |
| | Outcome Variable: 12th Grade Math Test Score | |
| | ATT Weight | ATC Weight |
| Model 3: Model 2 + expectations and parental involvement | .82 | 2.77 |
| | (.63) | (.74) |
| | Outcome Variable: Math Gain (12th – 10th) | |
| | ATT Weight | ATC Weight |
| Model 3: Model 2 + expectations and parental involvement | .98 | 1.94 |
| | (.29) | (.44) |

# Interpretation: Are the ATT and ATC both identified?

- We think it is reasonable to regard the ATT as identified because it is reasonable to assume that

$$E[Y^0|D = 1, S] = E[Y^0|D = 0, S]$$

  *and* for this example that $X$ is equivalent to $S$.
- Assumption 2-S is implied by a stronger assumption of partial ignorability with respect to $S$, such that

$$Y^0 \perp\!\!\!\perp D \,|\, S \tag{7.13}$$

- Nonetheless, it is possible that those students most likely to benefit from public schools are those students who are least likely to attend Catholic schools, net of $X$ (and contrary to Matching Demonstration 4 developed for Chapter 5).
- If so, Assumption 2-S does not obtain for the conditioning set $X$ because

$$E[Y^0|D = 1, X] < E[Y^0|D = 0, X]. \tag{7.14}$$

# Now the ATC ...

- Past research that maintains that self-selection is present
- We do not have any variables in the ELS that could plausibly measure students' (and their parents') own expectations for the benefits they might obtain from attending a Catholic school
- Accordingly, we cannot accept what is labeled Assumption 1-S in Equation (5.1),

$$E[Y^1|D = 1, S] = E[Y^1|D = 0, S],$$

or the stronger assumption of partial ignorability,

$$Y^1 \perp\!\!\!\perp D \,|\, S \tag{7.15}$$

because $X$ is *not* equivalent to $S$.

- We do not see any basis for regarding the ATC estimates as consistent or unbiased for the true ATC.
- Instead, these estimates are very likely too large because

$$E[Y^1|D = 1, X] > E[Y^1|D = 0, X]. \tag{7.16}$$

# Why might the true ATC still be larger than the true ATT?

The *common school* explanation: Catholic schools distribute opportunities for learning, such as advanced course-taking, more equitably than do public schools.

- Variables such as parental education and non-minority status have positive relationships with $D$ but negative relationships with the variation in $\delta$ among Catholic schools students.
- The explanation is based on the implicit claim that

$$E[Y^1 - Y^0 | D = 1, CSE = 1] > E[Y^1 - Y^0 | D = 1, CSE = 0] \tag{7.17}$$

  where $CSE$ is a "common school effect" dummy variable that indicates which particular students receive a "boost" in their individual-level treatment effect because of the comparatively egalitarian nature of Catholic schooling.

The *better alternatives* explanation: Catholic schooling is particularly beneficial to those students who have poor public schooling alternatives, in particular those students from families who are not able to afford to live in school districts with the best public schools.

- Variables such as family income and wealth have positive relationships with $D$ but negative relationships with the variation in $\delta$ among Catholic school students.
- The explanation is based on an implicit claim, of the same basic structure as Equation (7.17), that

$$E[Y^1 - Y^0 | D = 1, WAP = 1] > E[Y^1 - Y^0 | D = 1, WAP = 0] \tag{7.18}$$

where $WAP$ is a "worse alternative public school" indicator variable equal to 1 for those Catholic school students who have particularly poor public schooling alternatives, and which is assumed for this explanation to be a function in $S$.

- For the *better alternatives* explanation, the claim is implicitly in two separable pieces, which makes it qualitatively different than the common school explanation,

$$E[Y^1|D = 1, WAP = 1] = E[Y^1|D = 1, WAP = 0]$$

and

$$E[Y^0|D = 1, WAP = 1] < E[Y^0|D = 1, WAP = 0],$$

which together imply Equation (7.18).

- If either the common school explanation or the better alternatives explanation is true, then the true ATC would be larger than the true ATT, and this might also be true for the estimated ATC and ATT if the bias in the estimates is small enough not to obscure the true difference between the target parameters (as usual, assuming that sampling error is zero).

# Morgan's favored interpretation

- The *binding constraint* explanation: Differential responsiveness exists to accurate perceptions of students' likely benefits from Catholic schooling.
  1. Among low-income families for whom tuition at a Catholic school represents a genuine financial sacrifice, the students who enroll in Catholic schools are much more likely to be students who are likely to benefit from enrolling.
  2. Among high-income families for whom tuition is not a substantial financial sacrifice, even students who are not likely to benefit from attending Catholic schooling instead of public schooling may enroll in Catholic schools.

- Variables that capture resource availability have positive relationships with $D$ but negative relationships with the variation in $\delta$ among those who enter Catholic schooling.
- The implicit claim here is that

$$E[Y^1 - Y^0 | D = 1, BC = 1] > E[Y^1 - Y^0 | D = 1, BC = 0], \tag{7.19}$$

where $BC$ is an indicator variable equal to 1 for those students from families for whom Catholic school tuition is a genuine financial sacrifice, and which is therefore a function in $S$.

- For the *binding constraint* explanation, as with the better alternatives explanation, the claim is implicitly in two separable pieces,

$$E[Y^1|D = 1, BC = 1] > E[Y^1|D = 1, BC = 0]$$

and

$$E[Y^0|D = 1, BC = 1] = E[Y^0|D = 1, BC = 0],$$

which together imply Equation (7.19).

- In this case, when the average of individual-level treatment effects is weighted to the distribution of $S$ that characterizes the control group, the resulting average treatment effect is larger because this synthetic group of Catholic school students has more members with $BC = 1$.

# But perhaps a blended explanation is best

- The better alternatives and binding constraint explanations could both have merit, such that

$$E[Y^1|D=1, BC=1] > E[Y^1|D=1, BC=0]$$

and

$$E[Y^0|D=1, WAP=1] < E[Y^0|D=1, WAP=0],$$

with recognition that there is likely to be a strong positive association between $BC$ and $WAP$ in the population, conditional on $S$ (and its typical observed subset $X$).

- These assertions could then be combined with the claim that the difference between the estimated ATC and the estimated ATT is further increased because these assertions also lend indirect support to the claim that the estimated ATC is inconsistent and upwardly biased for the true ATC because

$$E[Y^1|D=1, X] > E[Y^1|D=0, X].$$

# Was it worth offering provisional estimates of the ATC?

- We knew from the outset which assumptions likely held, and it would have been reasonable to estimate and report only the ATT.
- Nonetheless, we think it is clear that what are most likely inconsistent and biased estimates of the ATC still give us a valuable perspective on why the ATC is not identified, and hence enhance our explanation of why, at best, only the ATT is identified by conditioning on the observed data.
- In addition, the reasoning that was motivated by the need to interpret the estimated ATC may also explain why the common school explanation of Coleman and his colleagues also may reflect underlying heterogeneity that is produced by individual-level selection on the treatment effect.